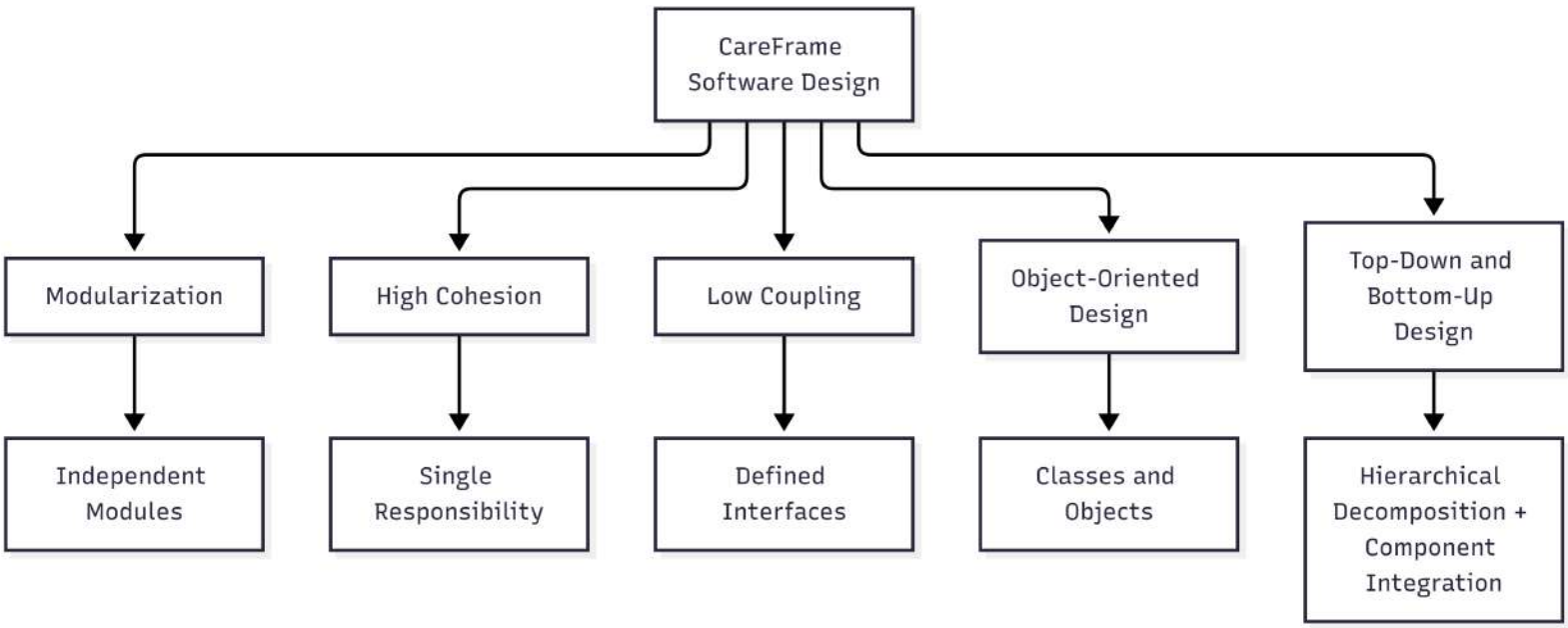
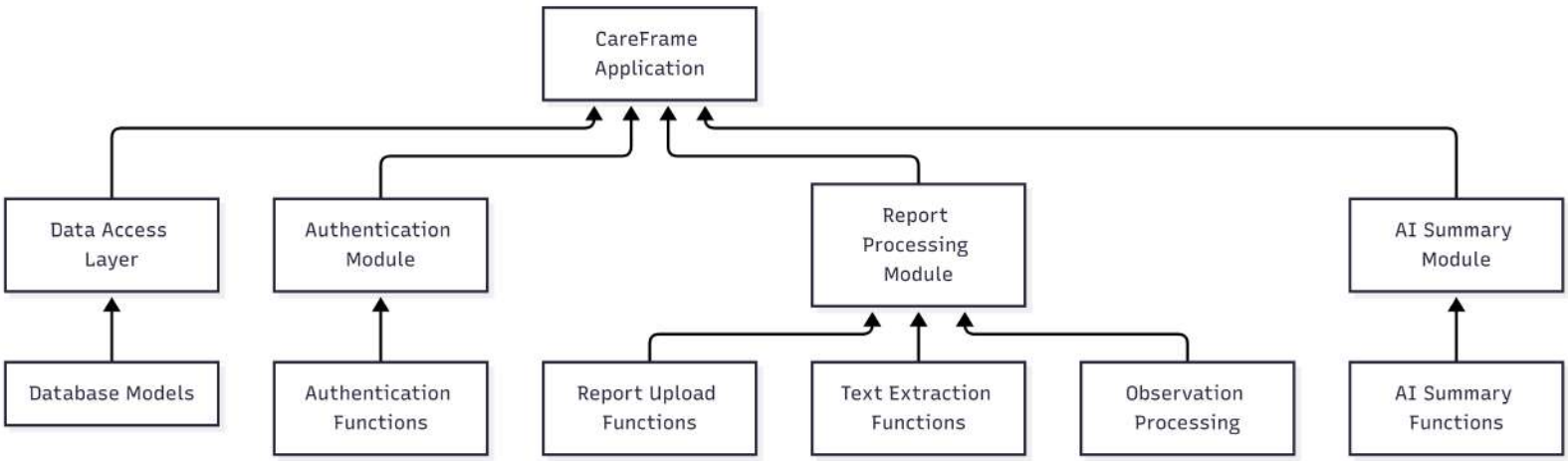
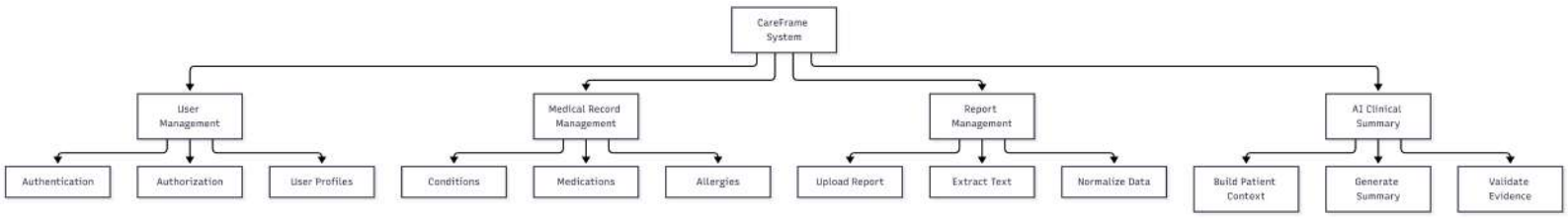
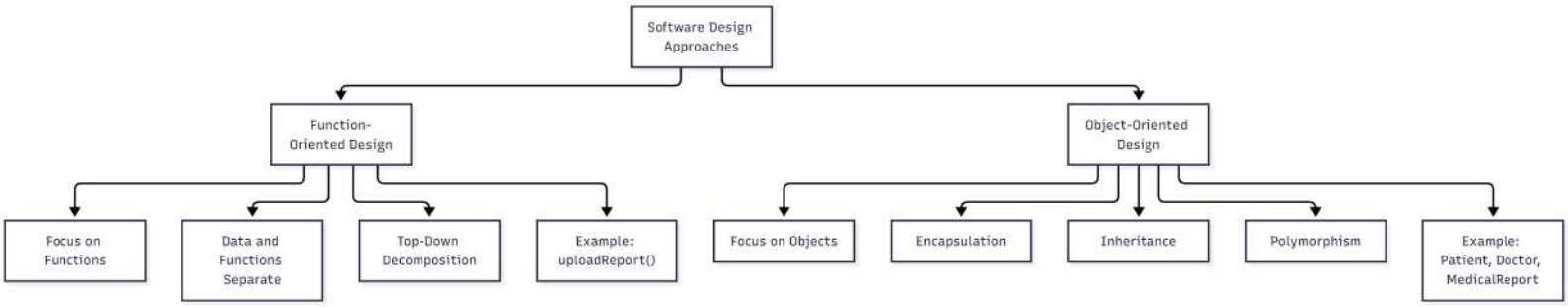
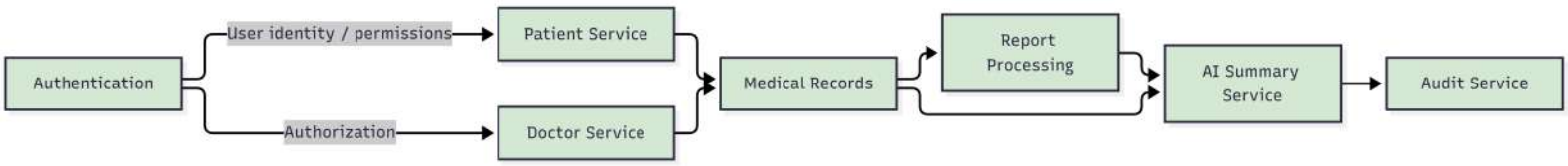
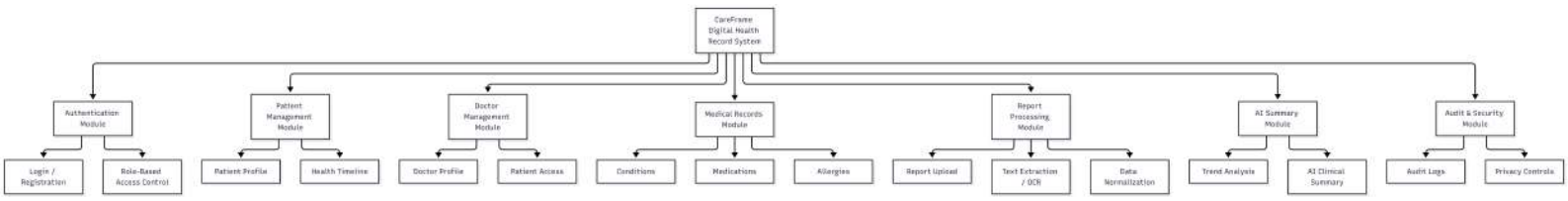

CareFrame
Software Design
Modularization
High Cohesion
Low Coupling
Object-Oriented
Design
Top-Down and
Bottom-Up
Design
Independent
Modules
Single
Responsibility
Defined
Interfaces
Classes and
Objects
Hierarchical
Decomposition +
Component
Integration


CareFrame Application
Data Access Layer
Authentication Module
Report Processing Module
AI Summary Module
Database Models
Authentication Functions
Report Upload Functions
Text Extraction Functions
Observation Processing
AI Summary Functions


CareFrame System
User Management
Authentication
Authorization
User Profiles
Medical Record Management
Conditions
Medications
Allergies
Report Management
Upload Report
Extract Text
Normalize Data
AI Clinical Summary
Build Patient Context
Generate Summary
Validate Evidence


Software Design Approaches
Function-Oriented Design
Object-Oriented Design
Focus on Functions
Data and Functions Separate
Top-Down Decomposition
Example: uploadReport()
Focus on Objects
Encapsulation
Inheritance
Polymorphism
Example: Patient, Doctor, MedicalReport


Authentication
User identity / permissions
Authorization
Patient Service
Doctor Service
Medical Records
Report Processing
AI Summary Service
Audit Service


CareFrame
Digital Health
Record System
Authentication
Module
Login /
Registration
Role-Based
Access Control
Patient
Management
Module
Patient Profile
Health Timeline
Doctor
Management
Module
Doctor Profile
Patient Access
Medical Records
Module
Conditions
Medications
Allergies
Report
Processing
Module
Report Upload
Text Extraction
/ OCR
Data
Normalization
AI Summary
Module
Trend Analysis
AI Clinical
Summary
Audit & Security
Module
Audit Logs
Privacy Controls